

through several years of prior education. Thus, the absence of switching effects in these occupations reinforces that the mobility effects we see elsewhere reflect a real link between chatbot use and newcomers gaining occupation-relevant skills.

**Discussion.** A consistent finding from experimental studies is that AI chatbots provide greater productivity benefits to workers with less prior expertise in their occupations (Brynjolfsson, Li and Raymond, 2025; Dell’Acqua et al., 2023; Noy and Zhang, 2023). This section presents large-scale evidence supporting a key implication of those studies: chatbot adoption is associated with occupational switching. In this sense, our findings align with the thesis of Autor (2024), who argues that Generative AI may help individuals extend their work into new occupations.

### 3.3 Workplace Employment

Figure 3 examines how employer chatbot initiatives affect workplace outcomes. Panel (a) first shows that workplaces that encourage chatbot usage have seen no differential change in total employment after the arrival of AI chatbots.<sup>16</sup> Our estimates are precise, ruling out effects larger than 2%.

Panel (b) focuses on the employment of early-career workers, which Brynjolfsson, Chandar and Chen (2025) document have seen a sharp drop in aggregate employment in the United States since 2022—a pattern Appendix G.2 replicates in the Danish labor market. Importantly, however, the difference-in-differences estimates in Figure 3, Panel (b) show that the decline in early-career jobs is not driven by firms adopting AI chatbots. Our estimates are again precise, ruling out effects on early-career employment greater than one-fifth of a full-time equivalent.

Figure 3, Panel (c) examines impacts on job churn. Even with constant employment levels, AI chatbots may require a different mix of skills, potentially increasing the rate of job creation and destruction in the workplace. For instance, adopting and operating

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<sup>16</sup>Appendix E.2.3 shows that wage bills are similarly flat. Furthermore, while Figure 3 focuses on employment within workers’ respective occupations, Appendix E.2.3 shows that overall workplace employment remains flat as well.

new tools may require hiring new workers who displace incumbents (Bessen et al., 2023), whereas assistance from the tools could instead help retain workers (Brynjolfsson, Li and Raymond, 2025). Yet rates of job creation and destruction have remained stable for workplaces that encourage chatbot use. Our confidence bands rule out effects on monthly job churn rates greater than 0.5% of employment.

Finally, even if overall rates of job churn are constant at the workplace, encouraging AI chatbots may attract workers who would like to use the tools. Figure 3, Panel (d) explores this possibility by estimating the “adopter propensity” of hires and separations at the workplace—that is, the predicted adoption rate (based on workers’ pre-determined characteristics) of workers who join or leave the workplace.<sup>17</sup> However, despite the model prediction that AI chatbots may lead to workers re-sorting across firms, we see no differential changes in the adopter propensity of workers who join or leave workplaces that encourage chatbot usage. The confidence bands rule out effects on the adopter propensities of hires and separations larger than 1 percentage point.

**Heterogeneity by Strength of Treatment.** Figure 4, Panels (c) and (d) examine the effects of additional initiatives in enterprise chatbots and training. Although Appendix D shows that uptake and reported benefits are highest when the full suite of initiatives is implemented, we find no detectable impacts on overall employment or early-career jobs at these workplaces. Similarly, Appendix E.2.3 shows that job churn and worker sorting have not changed differentially at workplaces with these investments.

**Occupational Heterogeneity.** Figure 5, Panels (c) and (d) report the estimated effects of AI chatbots on workplace employment and early-career jobs across the 11 occupations in our sample. We find no significant effects in any occupation: the estimates are close to zero and generally rule out effects larger than 10%.<sup>18</sup> Even between our most contrasting groups—software developers and marketers (with flexible labor rules) versus teachers

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<sup>17</sup>Appendix H.1 details the estimation of these adopter propensities, which our economic framework in Section A.2 microfound as workers’ latent willingness to use AI chatbots.

<sup>18</sup>Appendix E.2.3 similarly shows null results for job churn and worker sorting.

(with collective bargaining)—we observe no systematic differences in employment impacts. Moreover, while Appendix G.2 documents aggregate declines in early-career jobs in several of our exposed occupations, including software development, customer support, legal professions, and marketing, these declines are not driven by firms adopting Generative AI chatbots.

**Discussion.** In summary, workplaces that have adopted AI chatbots have not fared differently in terms of their employment, early-career jobs, job churn, and worker mix. These limited impacts contrast with the widespread creation of new job tasks revealed by our survey (Section 1.4). These patterns are consistent with the productivity J-curves often observed in the early phases of a new technology (Brynjolfsson, Li and Raymond, 2025; McElheran et al., 2025): while organizations have started to reorganize internally in response to AI chatbots, devoting substantial resources to integration, training, and oversight, these developments have not yet manifested in measurable changes in recorded hours or earnings.

## 4 Supplementary Evidence

### 4.1 Perceived Impacts

To complement the difference-in-differences analysis of labor market impacts, we simply asked respondents directly: *“Have AI chatbots affected your labor earnings?”* If so, we followed up with, *“By how much?”* If AI chatbots were meaningfully changing labor market conditions in exposed occupations, we might expect “front-line workers” in those occupations to notice and report such changes.

Appendix E.3 reports workers’ perceived earnings impacts from AI chatbots. Tellingly, virtually all respondents—97.7% of adopters and 99.5% of non-adopters—state that AI chatbots have had *no* impact on their labor earnings. While individuals may struggle to assess such counterfactuals precisely, these responses indicate three implications.

First, even workers themselves report small average earnings impacts, ranging from

0% to 0.5%, depending on their occupation and their employer’s policy on use. These estimates are an order of magnitude smaller than the time savings reported in Section 1.4 and fall within the confidence intervals of the observed earnings effects in Section 3.1.<sup>19</sup>

Second, the limited average earnings impacts primarily reflect the overwhelming share of workers reporting no change in earnings due to AI chatbots—rather than large positive and negative effects offsetting each other. This supports our conclusion from Figures 4 and 5 that AI chatbots have not had significant heterogeneous impacts on workers.

Finally, the fact that virtually 100% of non-adopters report no earnings impact suggests that occupation-wide spillovers are minimal. Despite broad adoption and changes in work content, AI chatbots do not appear to have shifted equilibrium wages or hours.

Taken together, these self-reported perceptions from front-line workers in the most exposed occupations reinforce our conclusion that AI chatbots have so far had limited impacts on labor markets.

## 5 Conclusion

Generative AI is heralded as the engine of a new industrial revolution (World Economic Forum, 2024), yet we lack evidence on its economic impacts outside laboratory settings and case studies (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023).

This paper provides large-scale and representative evidence on the early labor market impacts of AI chatbots, the most widely adopted Generative AI tool to date. Our study is based on a series of extensive surveys on AI chatbot usage in 11 exposed occupations, linked to administrative labor market data in Denmark.

Despite rapid adoption and substantial investments by both workers and firms, our key takeaway is that AI chatbots have had minimal impact on earnings and total hours to date. Moreover, we find no evidence of differential trends over time—even among “AI front-runners”—suggesting that the limited effects may persist.

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<sup>19</sup>As shown in Figure E.19, workers themselves report that only 3-7% of their time savings translate into higher earnings.